

INTERFERENCE

OPTICS IS THE SCIENTIFIC STUDY OF LIGHT

Optics is divided in to Ray optics and Wave optics

NEWTON'S CORPUSCULAR THEORY OF LIGHT

Light consists of stream of particles called corpuscles. They travel in a straight line with large velocities. The corpuscular theory was successful in explaining reflection, refraction of light and rectilinear propagations of light.

WAVE THEORY OF LIGHT

Light propagated through a medium in the form of transverse waves. Each point of a source of light is a centre of disturbance, from which waves are propagated in all directions. The Wave theory of light could explain almost all phenomena like reflection, refraction, interference, diffraction and polarisation

PERIOD(T)

The time taken by the particle to make one complete vibration is called period.

FREQUENCY(ν)

The number of vibrations made by the particle in one second is called frequency

Frequency $\nu = \frac{1}{T}$

WAVELENGTH(λ)

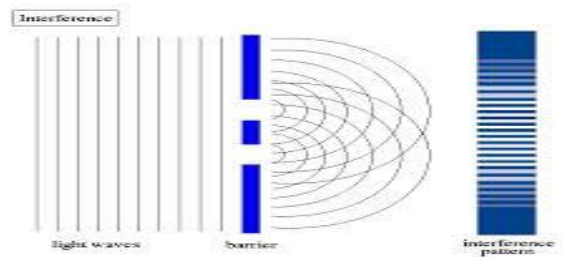
The distance between two successive crests or two successive trough is called wavelength

RELATION BETWEEN VELOCITY, WAVELENGTH AND FREQUENCY

$$V = \nu \lambda$$

INTERFERENCE

The remodification of light energy due to the superposition of two light wave of the same amplitude, same frequency and of constant phase difference is called interference



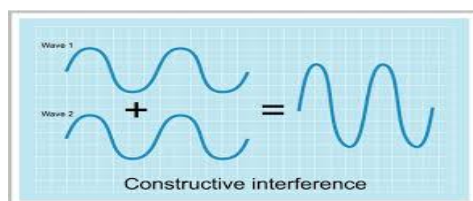
PRINCIPLE OF SUPERPOSITION

According to this principle, the resultant displacement of a particle of the medium acted on by two waves simultaneously, is equal to the algebraic sum of the displacement due to each wave

$$Y = y_1 + y_2$$

CONSTRUCTIVE INTERFERENCE

At the points in the region where the two light waves arrive in the same phase, the resultant intensity is maximum and the interference is said to be constructive



For Constructive Interference

The waves must arrive to the point of study in phase. So their path difference must be integral multiples of the wavelength:

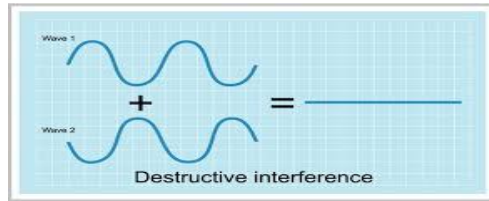
$$p.d = 0, 1\lambda, 2\lambda, 3\lambda, \dots$$

$$p.d = n\lambda$$

$$n = 0, 1, 2, 3, \dots$$

DESTRUCTIVE INTERFERENCE

At points where the two light waves arrive in phase opposition, the resultant intensity is minimum and the interference is said to be destructive



For destructive interference

The waves must arrive to the point of study out of phase. So the path difference must be an odd multiple of $\lambda/2$:

$$p.d = \lambda/2, 3\lambda/2, 5\lambda/2, \dots$$

$$p.d = (2n+1) \lambda/2 \quad n=0,1,2,\dots$$

COHERENT SOURCES

Two sources of light are said to be coherent if the waves emitted from them have the same frequency, same amplitude and zero or constant phase difference

EXAMPLES

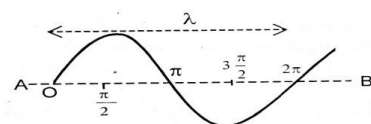
Two slits illuminated by a monochromatic source of light, a source of light and its reflected image

CONDITIONS FOR INTERFERENCE

- There should be two coherent source of light emitting light waves of same frequency and same amplitude with a constants phase difference
- The light waves from the coherent sources should superimpose, at the same time and at the same place
- The two coherent sources of light should be very close to each other
- The two sources must be very narrow
- The two sources should be of equal intensities

PHASE DIFFERENCE AND PATH DIFFERENCE

$$\text{Phase difference} = 2\pi/\lambda \times \text{path difference}$$



OPTICAL PATH

Optical path is the distance that light would travel in air or vacuum equivalent to the distance d it travels in a medium of refractive index μ

$$\text{OPTICAL PATH} = \mu d$$

Path difference due to dissimilar reflection

If a ray is reflected at a surface backed by a denser medium then there will be an additional path difference of $\lambda/2$

COHERENCE

The most important condition for obtaining a stable interference pattern is the coherence of the interfering waves. For this the waves should be continuous, sinusoidal and monochromatic.

The average length of a wave packet emitted by a source is called the coherence length, l_c .

The average time duration for this, is called the coherence time, t_c . $t_c = l_c/c$

An electromagnetic wave is associated with a varying electric field. The phase of the wave varies with time as well as position in space.

TEMPORAL COHERENCE / LONGITUDINAL COHERENCE

This refers to the coherence of the wave with respect to time. i.e., the wave will have constant phase change in equal intervals of time.

SPATIAL COHERENCE / TRANSVERSE COHERENCE

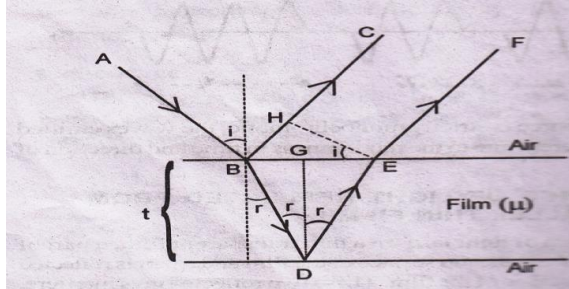
This refers to the coherence between the vibrations at two different points on a plane perpendicular to the direction of propagation of waves. Waves are said to have spatial coherence, if the phase difference between the vibrations at any two points on the plane remains constant.

INTERFERENCE OF LIGHT REFLECTED FROM PLANE PARALLEL THIN FILMS

When a beam of light falls on a thin film, a part of light is reflected from the top surface of the film and a part is reflected from the lower surface of the film. These two reflected rays interfere. If the incident light is white, the film is beautifully coloured.

BY REFLECTED LIGHT

Consider a thin transparent film of thickness t and refractive index μ . A ray of light AB is incident on the upper surface of the film. A part of the ray is reflected along BC . The remaining part is transmitted along BD .



Since the film is very thin the rays BC and EF are very close to each other hence they interfere and produce brightness or darkness according to the path difference between them. Draw EH perpendicular to BC. Then beyond EH, the two reflected rays travel the same distance.

The optical path difference = (BD + DE) in film – BH in air

$$\text{o.p.d} = (BD + DE) \mu - BH$$

ΔBDG and ΔEDG ARE congruent

$$BD = DE \text{ and } BG = GE$$

$$\text{o.p.d} = 2 BD \mu - BH \dots\dots\dots(1)$$

From ΔBEH

$$\sin i = BH/BE$$

$$BH = BE \sin i$$

$$BE = BG + GE \text{ and } BG = GE$$

$$BH = 2 BG \sin i$$

Substitute in eqn (1)

$$\text{o.p.d} = 2 BD \mu - 2 BG \sin i \dots\dots\dots(2)$$

From ΔBDG

$$\cos r = DG/BD = t/BD$$

$$BD = \frac{t}{\cos r} \dots\dots\dots(3)$$

$$\tan r = BG/t$$

$$BG = t \tan r \dots\dots\dots(4)$$

Substitute in eqn (3) and (4) in eqn (2)

$$\text{Opd} = \frac{2\mu t}{\cos r} - 2t \tan r \cdot \sin i$$

$$\begin{aligned}
 &= \frac{2\mu t}{\cos r} - 2t \frac{\sin r}{\cos r} \mu \sin r \\
 &= \frac{2\mu t}{\cos r} (1 - \sin^2 r) \\
 &= 2\mu t \cos r
 \end{aligned}$$

The reflection at B is at the surface of a denser medium. Hence reflected ray BC undergoes a phase change of π or the ray travels an additional distance of $\lambda/2$

Correct path difference = $2\mu t \cos r + \lambda/2$

CONDITION FOR CONSTRUCTIVE INTERFERENCE

Path difference = $n\lambda$
 $2\mu t \cos r + \lambda/2 = n\lambda$
 $2\mu t \cos r = (2n-1)\lambda/2$
 The film appears bright

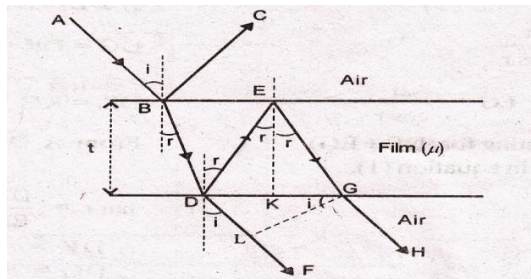
CONDITION FOR DESTRUCTIVE INTERFERENCE

Path difference = $(2n+1)\lambda/2$
 $2\mu t \cos r + \lambda/2 = (2n+1)\lambda/2$
 $2\mu t \cos r = n\lambda$
 The film appears dark

COSINE LAW

For destructive interference in the reflection, the condition is $2\mu t \cos r = n\lambda$ where $n = 1, 2, 3, \dots$
 This is cosine law.

INTERFERENCE IN THIN FILMS DUE TO TRANSMITTED LIGHT



Correct path difference = $2\mu t \cos r$

CONDITION FOR CONSTRUCTIVE INTERFERENCE

$2\mu t \cos r = n\lambda$
 The film appears bright

CONDITION FOR DESTRUCTIVE INTERFERENCE

$$2 \mu t \cos r = (2n+1) \lambda/2$$

The film appears dark

COLOURS OF THIN FILMS

A thin transparent film observed in white light appears coloured

Condition for darkness is $2 \mu t \cos r = n\lambda$

Depends on

Wavelength

Angle of refraction

Thickness

At any region of film, if μ , t and $\cos r$ satisfies this condition for particular wavelength λ , that colour will be absent in that reflected light. Hence that part of the film will appear to have the colours of remaining light.

It depends on angle of refraction r hence on angle of incidence i . So when we observe the same film from different positions the colour of the film will be varying.

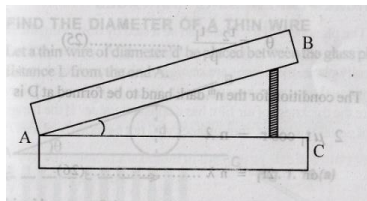
If the thickness is different, colour of the film will be varying. When t is very small, path difference between reflected rays $\lambda/2$, ie, darkness condition. Hence the film appears as dark.

When t is large, almost all colours undergo constructive interference. So the film appears to be white. So a thick film does not show colours.

ie, A thin film alone will show different colours.

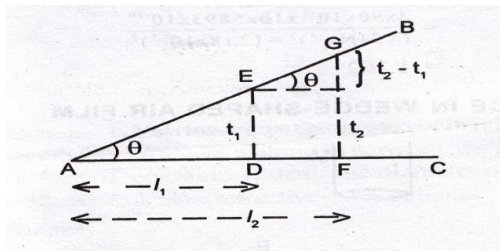
Eg: The brilliant colours of peacock feather, pigeon's neck etc

INTERFERENCE IN A WEDGE SHAPED FILM



Two plane glass plates are placed such that they are in contact at one end and separated by a small distance at the other end. A wedge shaped air film is formed in between them.

A beam of monochromatic light is incident normally on the lens. A part of the light is reflected from the top surface of the thin film another part of light is reflected from the top surface of the lower glass plate. Since thickness of air film is same along a line parallel to the line of contact of the glass plate, the interference pattern are parallel and straight to the line of contact. Each fringe is the locus of all points where the thickness of air film has a constant value. The angle between the glass plates is called ANGLE OF WEDGE



Let n th dark band is formed at D where thickness of thin film is t_1 and $(n+1)$ th dark band be formed at F where thickness of film is t_2 . Let $AD=l_1$ and $AF=l_2$

$$\tan \theta = (t_2 - t_1) / (l_2 - l_1)$$

Since θ is very small, $\tan \theta = \theta$

$$\theta = (t_2 - t_1) / (l_2 - l_1)$$

Condition for the n th dark band

$$2\mu t_1 \cos r = n\lambda$$

$$\mu = 1 \quad \& \quad \cos r = 1$$

$$2t_1 = n\lambda$$

For $(n+1)$ th dark band

$$2t_2 = (n+1)\lambda$$

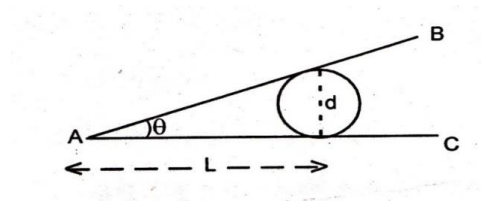
$$2(t_2 - t_1) = \lambda \quad (t_2 - t_1) = \lambda / 2$$

$$\text{Then} \quad \theta = \lambda / 2\beta$$

If the medium between the glass plate has refractive index μ

$$\theta = \lambda / 2\mu\beta$$

DIAMETER OF THE THIN WIRE



$$\theta = d/L \text{ and } \theta = \lambda / 2\beta$$

$$d/L = \lambda / 2\beta$$

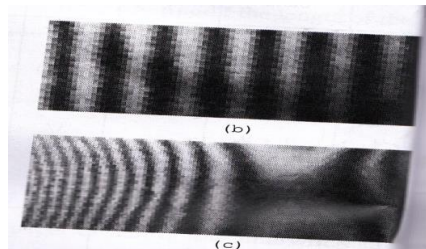
$$d = \frac{\lambda L}{2\beta}$$

OPTICAL PLANENESS OF SURFACES

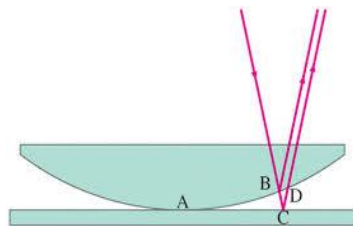
The planeness of a surface can be tested by observing the nature of fringes obtained using air wedge method.

If the fringes are straight and of equal thickness, the surfaces are optically plane

If the fringes are irregular (not straight) and not of equal thickness, the surfaces are not optically plane



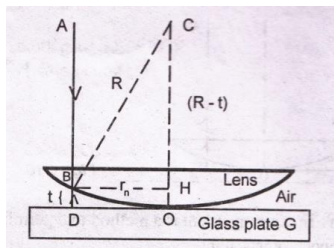
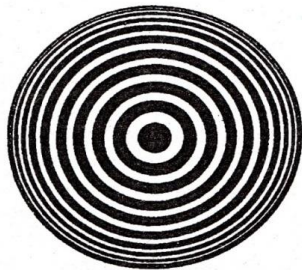
NEWTON'S RINGS



The arrangement consists of a plano convex lens of large radius of curvature placed on an optically plane glass plate. A thin film of air of varying thickness is formed between the lens and the glass plate. The thickness of thin film is zero at the point of contact and gradually increases towards the edge of the lens

A beam of monochromatic light is incident normally on the lens. A part of the light is reflected from the top surface of the thin film another part of light is reflected from the top surface of the glass plate. These two reflected beams interfere destructively or constructively and produce dark or bright band

If a point appears dark, all the points along a circle through this point are dark since the thickness of air film is same along a circle. So we get a dark ring. If the point appears bright, we get a bright ring. Thus alternate dark and bright rings of increasing radii are observed. As the radii increase, the rings become thinner and closer. These are called newton's rings.



At the point of contact between the lens and the glass plate, the thickness of air film $t=0$. Hence the path difference $2\mu t=0$. But at the time of reflection, the reflected wave from the glass plate undergoes a phase change of π (path difference of $\frac{\lambda}{2}$). So the waves at the centre are interfering destructively and hence a dark spot is obtained at the centre of newton's ring.

RADIUS OF THE n^{th} RING

Let C be the centre of curvature of the curved surface of the lens and R be its radius of curvature. Let r_n be the radius of dark ring, $BH=r_n$

The condition for the formation of the n^{th} dark ring at D is,

$$2\mu t \cos r = n\lambda \dots\dots\dots(1)$$

where $n = 0, 1, 2, 3 \dots\dots\dots$

Since the light is incident normally, $r=0$ or $\cos r = 1$.

Also $\mu = 1$ for air.

$$\therefore 2t = n\lambda \dots\dots\dots(2)$$

From the right angled triangle CBH,

$$CB^2 = CH^2 + BH^2$$

$$\text{ie, } R^2 = (R - t)^2 + r_n^2$$

$$= R^2 + t^2 - 2Rt + r_n^2.$$

Since t is very small t^2 is negligible.

$$\therefore R^2 = R^2 - 2Rt + r_n^2$$

$$\text{or } r_n^2 = 2Rt \dots\dots\dots(3)$$

$$2t = n\lambda \text{ from (2).}$$

$$\therefore r_n^2 = Rn\lambda \dots\dots\dots(4)$$

$$\text{or } r_n = \sqrt{Rn\lambda} \dots\dots\dots(5)$$

As n increases, the distance between the rings decreases.

RADIUS OF THE N^{th} BRIGHT RING

If the n^{th} bright ring is formed at B where the thickness of the air film is t , the condition for brightness is

$$2\mu t \cos r + \frac{\lambda}{2} = n\lambda \dots\dots\dots(6)$$

$$2\mu t + \frac{\lambda}{2} = n\lambda.$$

$\mu = 1$ for air, $\cos r = 1$ for normal incidence.

$$2 \mu t = (2n-1) \frac{\lambda}{2}.$$

$$\therefore 2t = (2n-1) \frac{\lambda}{2} \dots \dots \dots (7)$$

From equation (3), $r_n^2 = 2Rt$.

$$\therefore 2t = \frac{r_n^2}{R}$$

Substituting, $r_n^2 = R (2n-1) \frac{\lambda}{2}$

$$\text{or } r_n = \sqrt{R (2n-1) \frac{\lambda}{2}} \dots \dots \dots (8)$$

This is the radius of the n^{th} bright ring.

WAVELENGTH OF LIGHT

The radius of the n^{th} dark ring is

$$r_n = \sqrt{R n \lambda}$$

ie $r_n^2 = R n \lambda$

If D_n is the diameter of the n^{th} dark ring,

$$r_n = \frac{D_n}{2}$$

$$\text{or } r_n^2 = \frac{D_n^2}{4} = R n \lambda$$

$$\text{or } D_n^2 = 4R n \lambda \dots\dots\dots(9)$$

If D_{n+k} is the diameter of the $(n+k)^{\text{th}}$ dark ring,

$$D_{n+k}^2 = 4R (n+k) \lambda \dots\dots\dots(10)$$

$$\text{Then } D_{n+k}^2 - D_n^2 = 4R k \lambda.$$

$$\text{or } \lambda = \frac{D_{n+k}^2 - D_n^2}{4 k R} \dots\dots\dots(11)$$

Thus, measuring D_{n+k} , D_n , and R , the wavelength λ can be calculated.

Similarly, the radius of the n^{th} bright ring, r_n , is given by

$$r_n^2 = R(2n-1) \frac{\lambda}{2}$$

$$\text{Then } r_n = \frac{D_n}{2}$$

$$\text{or } r_n^2 = \frac{D_n^2}{4} = R(2n-1) \frac{\lambda}{2}$$

$$\text{or } D_n^2 = 4R(2n-1) \frac{\lambda}{2} \dots\dots\dots(12)$$

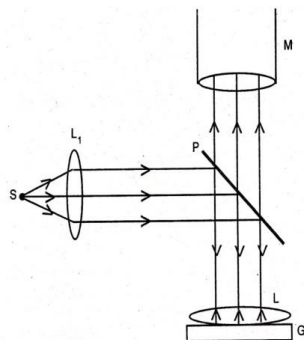
$$D_{n+k}^2 = 4R[2(n+k)-1] \frac{\lambda}{2} \dots\dots\dots(13)$$

$$D_{n+k}^2 - D_n^2 = 4Rk\lambda$$

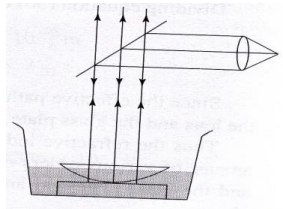
$$\lambda = [(D_{n+k})^2 - (D_n)^2] / 4 k R$$

WAVELENGTH OF LIGHT

$$\lambda = [(D_{n+k})^2 - (D_n)^2] / 4 k R$$



REFRACTIVE INDEX OF LIQUID



Let a liquid of refractive index μ be introduced between the glass plate and the lens. The radius of the n^{th} dark ring is given by

$$r_n^2 = 2Rt$$

$$\text{or } 2t = \frac{r_n^2}{R} \dots\dots\dots(17)$$

where 't' is the thickness of the liquid film where the n^{th} dark ring is formed and R is the radius of curvature of the lens.

The condition for the formation of the n^{th} dark ring is

$$2\mu t \cos r = n\lambda$$

Since $\cos r = 1$ for normal incidence.

$$\text{Then } 2t = \frac{n\lambda}{\mu} \dots\dots\dots(18)$$

$$\frac{r_n^2}{R} = \frac{n \lambda}{\mu}$$

or $r_n^2 = \frac{n R \lambda}{\mu} \dots\dots\dots(19)$

If d_n is the diameter of the n^{th} dark ring,

$$r_n = \frac{d_n}{2}$$

$$\therefore \frac{d_n^2}{4} = \frac{n R \lambda}{\mu}$$

or $d_n^2 = \frac{4 n R \lambda}{\mu} \dots\dots\dots(20)$

If d_{n+k} is the diameter of the $(n+k)^{\text{th}}$ dark ring,

$$d_{n+k}^2 = \frac{4 (n+k) R \lambda}{\mu} \dots\dots\dots(21)$$

$$\therefore d_{n+k}^2 - d_n^2 = \frac{4 k R \lambda}{\mu} \dots\dots\dots(22)$$

With air between the lens and the glass plate, we have,

$$D_{n+k}^2 - D_n^2 = 4 k R \lambda \dots\dots\dots(23)$$

The refractive index of the liquid, $\mu = \frac{(Dn+k^2-Dn^2)}{(dn+k^2-dn^2)}$

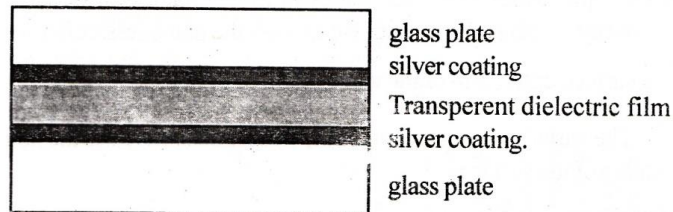
It can be shown that $\mu = \frac{r_1^2}{r_2^2} = \frac{D^2}{d^2}$ Since $\mu > 1$, $r_1 > r_2$. So the rings are contracting when a liquid is introduced. Where r_1 =radius of n^{th} ring with air and r_2 =radius of the same n^{th} ring with liquid as the medium.

INTERFERENCE FILTER

This is a very sensitive device used to filter out a monochromatic beam as a narrow band from the incident white light. When a parallel and narrow beam of white light is incident normally on a thin film the transmitted light will contain several bright bands. The condition for maxima at transmission is $2\mu t = n\lambda$

The wavelength of transmitted beam $\lambda = 2\mu t/n$

If t is very small, the no. of maxima can be made one or two. If t is large, a large number of maxima will be obtained in the transmitted light. The arrangement for filtering out a monochromatic beam from the incident white light is called interference filter

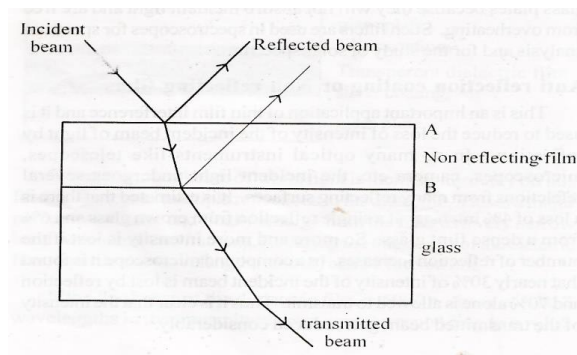


Interference filter consists of a very thin film of transparent dielectric material like magnesium flouride placed in between two optically plane glass plate. A very thin metallic coating of silver is deposited on the inner surface of glass plate.

When a narrow beam of white light is incident normally on this filter, it undergoes multiple reflections in the dielectric, it loses a major part of its intensity and finally emerge from the filter as a narrow beam with one or two maxima. Reducing the thickness t of the dielectric film, a particular wavelength can be filtered out.

ANTI REFLECTION COATING

This is an important application of thin film interference and it is used to reduce the loss of intensity of the incident beam of light by reflection



The loss of intensity due to reflection can be reduced by coating the reflecting surface with a suitable transparent dielectric material such as calcium flouride. The refractive index of such material must be in between that of air and glass. Such a film is called non reflecting film.

When a narrow and parallel beam of white light is incident on this film, a part of it is reflected from the upper surface A and lower surface B of the film. Now the thickness of the film is so adjusted that the reflections from A and B are in opposite phase and they cancel each other by destructive interference. Thus the loss of intensity during reflection is reduced and the beam is transmitted with maximum intensity.

The condition for destructive interference is $2\mu t = \lambda/2$

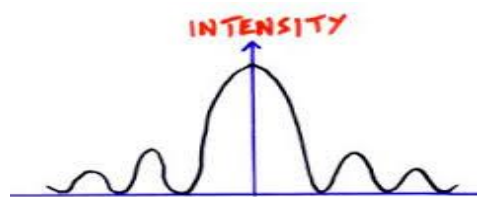
The thickness of film $t = \lambda/4\mu$

DIFFRACTION

Diffraction of light is the phenomenon of bending of light round the edges of an obstacle OR encroachment of light into the geometrical shadow of the obstacle.

DIFFRACTION PATTERNS

The intensity distribution on the screen is known as diffraction pattern



WAVEFRONT

A wavefront is an imaginary line that connects waves that are moving in phase

HUYGEN'S PRINCIPLE

Huygens principle states that each point on the wavefront will become a source of secondary waves spreading wavelets in all direction.

Diffraction is due to the mutual interference of the secondary wavelets originating from various points of the wavefront

COMPARISON BETWEEN INTERFERENCE AND DIFFRACTION

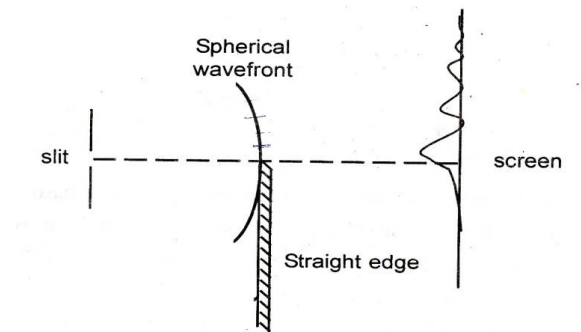
Interference	Diffraction
Interference bands are formed by superposition of waves from two coherent sources	Diffraction bands are formed by superposition of waves from different parts of the same wavefront
Bands are of equal width	Bands are of unequal width
Bands of minimum intensity are almost dark	Bands of minimum intensity are not dark
Intensity of bright bands is same	Intensity of bright bands is not the same

TYPES OF DIFFRACTION

Based on the position of the source of light and the screen with respect to the obstacle causing diffraction, there are two classes of diffraction, namely, Fresnel Diffraction and Fraunhofer Diffraction

FRESNEL DIFFRACTION

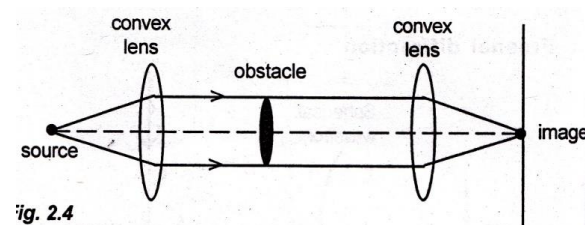
Either the source of light or the screen or both are at finite distance from the obstacle causing diffraction. Wavefront falling on the obstacle is spherical. Lenses are not used.



Example :Diffraction at a straight edge

FRAUNHOFER DIFFRACTION

The source of light and the screen are at infinite distance with respect to the obstacle causing diffraction. Wavefront falling on the obstacle is plane. Lenses are used.

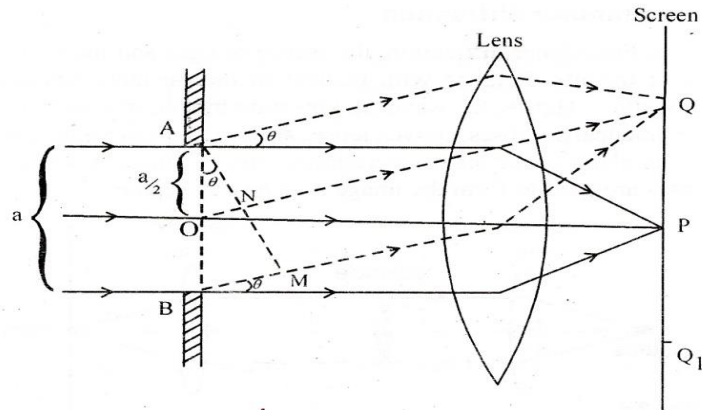


Example:Diffraction at a grating

FRAUNHOFER DIFFRACTION AT A SINGLE SLIT

Let AB be the narrow slit of width a . A plane wavefront of monochromatic light of wavelength λ is falling normally on this slit.

According to Huygens principle each point on the wave front behaves like secondary source producing secondary waves. O is the centre of slit. The straight and parallel waves from each point can be focused on the screen using lens. These straight waves path difference will be zero. They will interfere constructively producing brightness at the centre. This central bright band is called zero order central maximum.



Consider a point Q on the screen just above P. Here waves are reaching at an angle θ with the direction of OP. AM is the perpendicular drawn. Beyond AM, the waves have the same path. So BM gives the path difference.

From ΔABM $BM = AB \sin \theta$

$BM = a \sin \theta$

Let $BM = \lambda$, the wavelength of light. $\lambda = a \sin \theta$

The point Q will be bright or dark depending on the path difference between the waves reaching there. Let us consider that the slit may be divided into two halves AO and OB

$AO = OB = a/2$

$ON = a/2 \sin \theta$ and $BM = a \sin \theta$

If this path difference is $\lambda/2$, these two waves will interfere destructively and produce minimum intensity at Q

$a/2 \sin \theta = \lambda/2$

$\sin \theta = \lambda/a$

so the intensity at Q will be zero and the dark band at Q is called first order minimum.

Second, third etc order will be obtained if $\sin \theta = 2\lambda/a, 3\lambda/a, 4\lambda/a$ etc

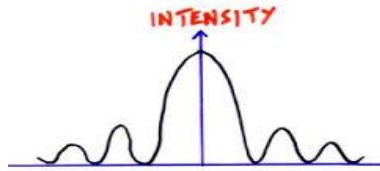
general condition for minima is given as

$\sin \theta_n = n\lambda/a$

general condition for maxima is given as

$\sin \theta_n = (2n+1) \lambda/2a$

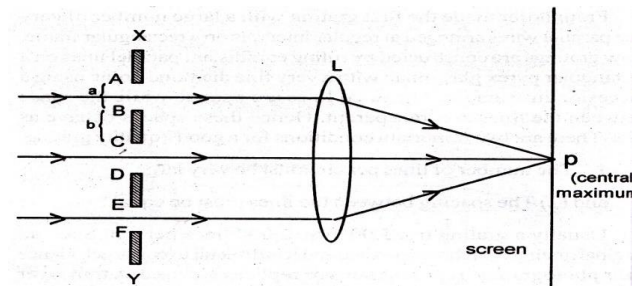
secondary maxima will be obtained for the angles $\theta_n = 3\lambda/2a, 5\lambda/2a$ etc



PLANE TRANSMISSION GRATING

Grating is an arrangement of a large number of parallel slits of equal width separated by equal opaque spaces

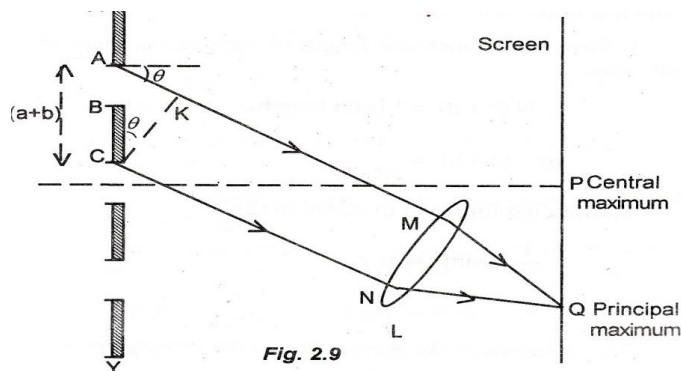
PLANE TRANSMISSION GRATING is a plane glass plate containing a large number of equidistant parallel lines drawn using a fine diamond point. The space between the lines acts as narrow slits through which light is transmitted. The lines are opaque to light.



Let a be the width of slit and b be the width of line the distance $a+b$ is called grating element. A plane wavefront of monochromatic light of wavelength λ is falling normally on this slit.

Each point of the wave front sends out secondary waves in all direction. The straight and parallel waves from each point can be focused on the screen using lens. These straight waves path difference will be zero. They will interfere constructively producing brightness at the centre. This central bright band is called central maximum.

The position of central maximum is same for all the wavelength. The central maxima will have the same colour as the incident light. Consider two waves diffracted from two points A and C of slit. Then the path difference between the two waves is AK .



From triangle ACK

$$\sin \theta = AK/AC$$

$$AK = AC \sin \theta = (a+b) \sin \theta$$

If $(a+b) \sin \theta = n\lambda$ two waves interfere constructively.

If $n=1$, it is the first order principal maximum, if $n=2$, it is the second order principal maximum and so on.

Thus on either side of central maximum, a number of principal maxima are obtained.

If there are N lines/unit length of the grating, there are N slit also.

$$N(a+b) = 1$$

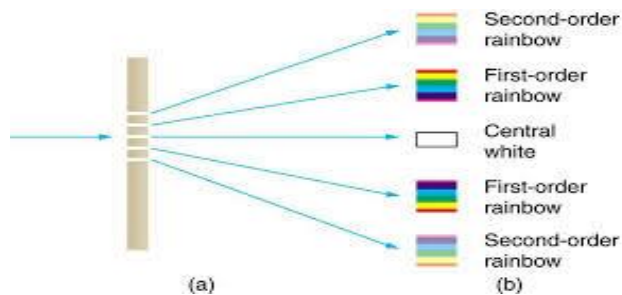
$$(a+b) = 1/N$$

$$1/N \sin \theta = n\lambda$$

$$\sin \theta = Nn\lambda$$

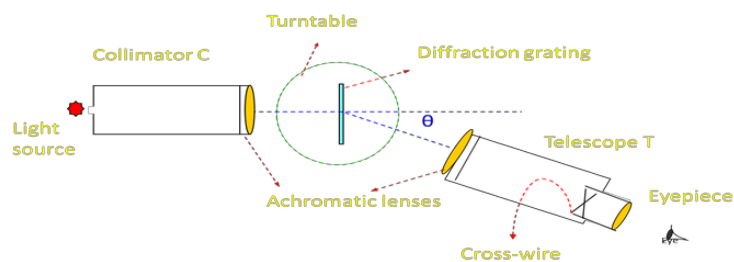
This is known as grating law.

For a grating θ is different for different colours (λ), for each value of n . if white light is used, it gets split up into different colours.



Using a diffraction grating to measure the wavelength of light

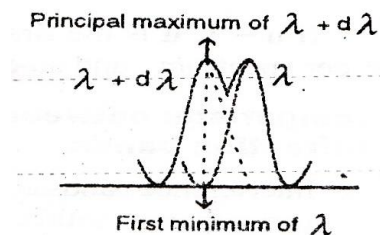
A spectrometer is a device to measure wavelengths of light accurately using diffraction grating to separate.



Study the expt: from lab manual.

RAYLEIGH'S CRITERION FOR RESOLUTION OF SPECTRAL LINES

According to Rayleigh's criterion for resolution, two neighbouring spectral lines will be just resolved when the principal maximum of one in any order falls on the first minimum of the other in the same order



Let λ and $\lambda + d\lambda$ be the wavelengths corresponding to two neighbouring spectral lines of the same order. Then, the two spectral lines are visible as separate when the principal maximum of wavelength $\lambda + d\lambda$ falls on the first minimum of wavelength λ of the same order.

RESOLVING POWER OF GRATING

It is the ability to show two neighboring spectral lines in a spectrum as separate. If λ and $\lambda + d\lambda$ are the wavelengths of two neighbouring spectral lines, the resolving power of the grating is defined as the ratio,

$$\lambda / d\lambda = nN_1 N_1 \text{--- total number of lines}$$

According to Rayleigh's criterion for resolution, two neighbouring spectral lines will be just resolved when the principal maximum n th order for $\lambda + d\lambda$ falls on the first minimum of the same order for λ . This is possible if the extra path difference introduced is $\frac{\lambda}{N_1}$. The angle of diffraction for both is same

The condition for the first minimum of n th order for λ

$$(a+b) \sin \theta = n\lambda + \frac{\lambda}{N_1} \text{-----(1)}$$

The principal maximum for a wavelength $\lambda + d\lambda$ is given by

$$(a+b) \sin \theta = n(\lambda + d\lambda) \text{-----(2)}$$

equating RHS of eqn (1) & (2),

$$\begin{aligned} n(\lambda + d\lambda) &= n\lambda + \frac{\lambda}{N_1} \\ nd\lambda &= \frac{\lambda}{N_1} \end{aligned}$$

The resolving power of the grating,

$$\frac{\lambda}{d\lambda} = nN_1$$

It is proportional to the order n and total number of lines N_1 on the Grating.

DISPERSIVE POWER OF A GRATING

The dispersive power of a grating is defined as the ratio of the change in the angle of diffraction to the corresponding change in wavelength.

Let two wavelengths λ and $\lambda + d\lambda$ be diffracted through angles θ and $\theta + d\theta$ respectively. Condition for n^{th} order maximum for a wavelength λ is

$$(a+b) \sin \theta = n\lambda \text{ -----(1)}$$

Differentiating, $(a+b) \cos \theta d\theta = n d\lambda$

$$\frac{d\theta}{d\lambda} = \frac{n}{(a+b) \cos \theta}$$

But $\frac{1}{a+b} = N$, the no of lines /unit length.

$$\text{Dispersive Power, } \frac{d\theta}{d\lambda} = \frac{Nn}{\cos \theta}$$

For small values $\cos \theta = 1$

Such a spectrum is called a normal spectrum. It is proportional to the order n and the number of lines per unit length N .

Uses of grating

1. Determination of wavelength
2. Spectral resolution difference in wavelength of two closely spaced spectral lines.
3. Spectral composition of light source (dispersion).